llmXive Automated Review of $ $-Bench: Evaluating Proactive Personal Assistant Agents in Long-Horizon Workflows

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The manuscript presents a novel benchmark for proactive agents, but several factual claims and citations require verification to ensure accuracy.

First, the bibliography and citation list contain significant anomalies. The provided proofreader flags indicate a mismatch for `c-001` (DOI vs. URL). More critically, the paper evaluates models such as "GPT-5.4", "Gemini-3.1 Pro", and "Claude 4.6 Opus". As of the current date, these specific model versions have not been publicly released. If these are hypothetical models used for simulation, the paper must explicitly state this to avoid misleading readers into believing these are real-world performance benchmarks. If they are internal or future-dated models, the claims of their specific scores (e.g., GPT-5.4 achieving 67.0% Proactivity) are currently unverifiable by the community.

Second, there is a discrepancy between the textual claims in Section 4.2 ("Analysis") and the data presented in Table 1 ("Task Type Scores"). The text states: "legal matter operations (H) have high Comp (84.1%) but low Proc (38.1%); drug design (K) shows the opposite (84.9% vs. 68.0%)." However, inspecting Table 1 (Task Type Scores) reveals:

- For Type H (Legal Matter Operations), the GPT-5.4 row shows 39/85 (Proc/Comp). The text's 84.1% Comp figure does not match the 85% shown for GPT-5.4, nor does the 38.1% Proc match the 39%.
- For Type K (Drug Design), the text claims 84.9% Proc and 68.0% Comp. The table shows GPT-5.4 at 78/74. The text's numbers appear to be aggregate averages across all models, but the table does not provide a "Total" or "Average" row for these specific types, making the claim difficult to verify against the provided visual data. The text should either provide the source of these aggregate numbers or correct the values to match the specific model data shown if the claim is intended to be about the leading model.

Finally, the citation keys (e.g., `openclaw`, `nanobot`, `gpt54`) are not defined in the provided bibliography snippet. While external availability is acceptable, the specific mapping of these keys to real, accessible papers or repositories must be accurate. The presence of future-dated citations (e.g., `2026 in kim2026persona2web`) suggests the paper may be a preprint from a future timeline or a simulation; if the latter, the "Experiments" section must clarify that the results are synthetic or simulated, not empirical measurements of existing systems.

Action items.

- [**writing**] The bibliography contains a verified URL mismatch for citation c-001 (PyTorch download link vs. DOI 10.1145/345678.901234). Verify that all cited works (e.g., openclaw, nanobot, gpt54) correspond to actual, accessible sources, as several model names (GPT-5.4, Gemini-3.1) appear to be future-dated or hypothetical.
- [**writing**] Section 4.2 claims 'legal matter operations (H) have high Comp (84.1%) but low Proc (38.1%)'. However, Table 1 (Task Type Scores) shows Type H scores for GPT-5.4 as 39/85 and Type K (Drug Design) as 78/74. The text's specific aggregate numbers (84.1/38.1) do not match the visible row data in Table 1 or the text description of Type K (84.9/68.0). Clarify if these are weighted averages not shown in the table or correct the values to match

the table.

- **[science]** The paper cites ‘GPT-5.4’, ‘Gemini-3.1 Pro’, and ‘Claude 4.6 Opus’ as evaluated models. These model versions do not currently exist in public release (as of the paper’s apparent context). If these are hypothetical or internal models, the claims regarding their specific performance scores (e.g., 67.0% Proc) are factually unverifiable and potentially misleading without explicit clarification of their provenance.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

The figure provides a clear visual overview of the benchmark’s components, but the caption is grammatically incomplete, and the diagram lacks a legend to explain the specific dependency types represented by the colored arrows in the episode structure.

Figure 2

The figure provides a clear visual overview of the benchmark session loop and intent tracking, but the caption is defective due to missing text where the three terminal status categories should be listed.

Figure 3

The figure effectively visualizes the relationship between Proactivity and Completeness across different task types with a clear legend. However, the caption contains typos and incomplete cross-references that need correction.

Action items.

- **[fatal]** Figure 1: The caption reads ‘Overview of .’ which is incomplete and missing the benchmark name (likely ‘ -Bench’).
- **[science]** Figure 1: Panel (b) ‘Episode Structure’ shows a timeline of sessions (S1-S20) but lacks a clear legend defining the specific meaning of the different arrow colors (blue, purple, green) connecting the sessions.
- **[writing]** Figure 2: The caption contains missing text for the terminal status categories (‘as , , or .’); the figure shows ‘completed’, ‘inferred’, and ‘provided’, so the caption must be

updated to include these terms.

- [**writing**] Figure 3: The caption contains typos and placeholders, specifically '(bluea, b, c)' and 'Tab. of App. ,', which makes the reference to the taxonomy table and appendix incomplete.
- [**writing**] Figure 3: The axis labels 'Proactivity' and 'Completeness' do not match the caption's abbreviations 'Comp' and 'Proc', creating a minor inconsistency in terminology.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology that creates a barrier for non-specialist readers, particularly in the Abstract, Introduction, and Evaluation Protocol sections.

First, the core concept of **"hidden intents"** is introduced without a plain-language definition. The text describes them as "habits, constraints, preferences" but relies on the jargon "underspecification" to explain the problem. A general reader needs a sentence explaining that this means "requirements the user forgot to mention or didn't think to state." Similarly, **"proactivity"** is defined via a citation and abstract concepts like "anticipate user needs." The authors should provide a concrete, jargon-free example (e.g., "an agent that notices you usually order coffee at 9 AM and asks if you want one before you ask") before using the formal definition.

Second, the metrics **"Proc"** and **"Comp"** are introduced as acronyms but then used as standalone nouns throughout the text (e.g., "average Proc spans 43.1–67.0%"). This is a common academic shorthand that alienates readers unfamiliar with the specific notation. The full terms "Proactivity Score" and "Completeness Score" should be used at least once in every paragraph where they appear, or the acronyms should be defined more prominently.

Third, terms like **"terminal status"** (Section 3.1) and **"scaffold"** (Section 4.1) are used without explanation. "Terminal status" is a state-machine term; "final resolution state" is clearer. "Scaffold" is engineering jargon for the underlying code framework; "framework" or "architecture" is more accessible.

Finally, the phrase **"long-horizon"** is used repeatedly. While standard in Reinforcement Learning, it is opaque to generalists. "Long-term" or "multi-step" would be more inclusive. The paper would benefit from a "Glossary of Terms" or a dedicated paragraph in the Introduction translating these technical concepts into plain English.

Action items.

- [**writing**] The manuscript relies heavily on specialized terminology that creates a barrier for non-specialist readers, particularly in the Abstract, Introduction, and Evaluation Protocol sections. First, the core concept of "hidden intents" is introduced without a plain-language definition. The text describes them as "habits, constraints, preferences" but relies on the jargon "underspecification" to explain the problem. A general reader needs a sentence explaining that this means "requirements the user for

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The logical consistency of the paper is generally strong in its separation of Proactivity (Proc) and Completeness (Comp). The definitions are mutually exclusive and cover the full space of intent resolution. However, there are two areas where the causal claims and metric interpretations require tighter logical alignment.

First, the claim that "proactive assistance reduces user burden" (Abstract, Conclusions) is logically strained by the definition of the `inferred` status. The metric counts an intent as proactive if the agent asks a question and the user answers it. This still requires user effort (answering the question). The paper argues that high-proactivity models (like GPT-5.4) have fewer turns (Fig. 4), but if `inferred` counts as proactive, the reduction in turns must be driven almost entirely by the `completed` status (where the agent acts without asking). The paper does not explicitly disentangle the contribution of `completed` vs. `inferred` to the turn-count reduction. If `inferred` tasks still require user turns, the claim that proactivity *reduces* burden is only partially supported; it shifts the burden from "providing requirements" to "answering clarifications." The logic holds that proactivity is distinct from completeness, but the "reduced burden" conclusion needs to be qualified to reflect that it applies primarily to `completed` intents, not `inferred` ones.

Second, the ablation study (Fig. 5) presents a causal claim that seems counter-intuitive given the task design. The study removes prior sessions to test the value of history. The result shows a large drop in Proactivity (-9.5) but a small drop in Completeness (~2.5). Logically, if hidden intents are "latent requirements" essential for the task (as defined in Sec. 3.1), and the agent fails to resolve them (low Proc) due to missing history, the agent should fail to meet the checklist criteria (low Comp). The small delta in Completeness suggests that either: (a) the checklist criteria are not sensitive to the hidden intents (contradicting the definition of hidden intents as "affecting task handling"), or (b) the agent can still satisfy the checklist by guessing or by the user eventually providing the info (which would be `provided` status, not `completed/inferred`). The paper does not explain why the failure to be proactive (due to missing history) does not cascade into a significant failure in task completion. This weakens the causal link between "prior interaction" and "task success" as measured by the checklist.

Finally, the distinction between `inferred` and `provided` relies on the quality of the agent's question ("targeted" vs. "generic"). The case studies show agents asking questions that result in `provided` status (e.g., Kimi asking for the theme). The logic for why a specific question leads to `inferred` while another leads to `provided` is not formalized in the text, leaving a gap in the reproducibility of the metric. If the user agent's judgment of "targeted" is subjective, the logical consistency of the `inferred` category is compromised.

Action items.

- [**science**] The claim that proactivity reduces user burden is logically strained because 'inferred' status (agent asks, user answers) still requires user effort. Clarify if the turn-count reduction is driven solely by 'completed' intents, or if 'inferred' truly reduces burden compared to 'provided'.
- [**science**] The ablation study shows a large Proactivity drop but small Completeness drop when history is removed. Logically, if hidden intents are essential, missing them should fail the checklist. Explain why the lack of context does not cascade into significant Completeness failures.
- [**writing**] The distinction between 'inferred' (targeted question) and 'provided' (generic question) is not formally defined. Without clear criteria for what makes a question 'targeted', the status assignment logic is subjective and threatens metric reproducibility.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several strong claims regarding the nature of proactive agents and the generalizability of its findings that extend slightly beyond the strict bounds of the provided data and experimental scope.

First, the abstract and conclusion assert that "proactive assistance remains challenging" as a general finding. While the benchmark scores (ranging 43.1–67.0% for Proactivity) certainly indicate difficulty, the experimental setup is constrained to a single "Nanobot-style scaffold" and simulated user agents (Sec. 6, Limitations). The paper extrapolates these results to the broader field of "personal assistant agents" without sufficient qualification. The observed challenges may be artifacts of the specific scaffold's limitations in handling long-horizon state or the specific design of the simulated users, rather than an inherent limitation of the models themselves. The language should be tempered to reflect that these challenges are observed within the specific constraints of the benchmark environment.

Second, the claim of a "clear distinction" between task completion and proactivity (Abstract, Sec. 4.2) is supported by the data showing divergent scores in specific domains (e.g., Legal tasks having high Comp but low Proc). However, the paper overgeneralizes this into a fundamental separation of capabilities. The data also shows models like Qwen3.6 Plus achieving competitive scores on both metrics simultaneously, and the "distinction" appears highly dependent on the task type (e.g., Drug Design vs. Legal Operations). The text implies a universal decoupling that the data only supports as a conditional phenomenon. The discussion should clarify that this distinction is task-dependent and not necessarily a universal property of agent architectures.

Finally, the ablation study (Fig. 5) concludes that removing prior sessions "confirms that history aids proactive intent resolution." While the drop in Proactivity scores is evident, the paper overclaims the *mechanism* of this drop. The removal of prior sessions eliminates not just "intent context" but also explicit workspace state, tool history, and artifact continuity. Attributing the performance drop solely to the loss of "proactive intent resolution" ignores the possibility that agents simply failed due to missing explicit state information required for the task, rather than a failure of proactivity. The analysis should avoid conflating the loss of context with the loss of proactivity specifically.

Action items.

- [**writing**] The paper makes several strong claims regarding the nature of proactive agents and the generalizability of its findings that extend slightly beyond the strict bounds of the provided data and experimental scope. First, the abstract and conclusion assert that "proactive assistance remains challenging" as a general finding. While the benchmark scores (ranging 43.1–67.0% for Proactivity) certainly indicate difficulty, the experimental setup is constrained to a single "Nanobot-style scaffold" and sim

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The manuscript addresses a critical area in AI safety: the evaluation of proactive agents that infer hidden user intents. The separation of "Proactivity" (intent resolution) from "Completeness" (task success) is a valuable contribution for diagnosing agent behavior. However, from a safety and ethics perspective, there are significant concerns regarding the evaluation methodology and the handling of high-stakes domains.

First, the evaluation of "hidden intents" relies entirely on a simulated user agent (GPT-5.4) to determine if an agent's action was proactive, inferred, or if the user had to provide the information (Sec. 3.2, App. 4.1). This introduces a potential circularity and bias: the "ground truth" of what constitutes a valid hidden intent or a successful inference is determined by another LLM. If the evaluator model has specific biases regarding what constitutes "proactive" behavior, the benchmark scores may reflect the evaluator's preferences rather than genuine alignment with human needs. The authors should explicitly discuss this limitation and the potential for model bias in the intent resolution process.

Second, the benchmark includes tasks in high-stakes domains such as law (legal drafting, evidence strategy), finance (model validation), and pharmacy (drug design, clinical evidence) (Table 1, Appendix 2). While the authors note that the data is synthetic and sanitized, the paper currently lacks a robust disclaimer regarding the deployment of such agents. There is a risk that readers or downstream developers might interpret the benchmark results as a green light for deploying these agents in real-world legal or medical workflows. The “Societal Impacts” section (Appendix 5) is brief; it should be expanded to explicitly warn against the use of these agents in high-stakes decision-making without rigorous human-in-the-loop oversight and domain-specific validation.

Finally, the use of LLMs (GPT-5.4) as the primary rubric grader for checklist criteria (Appendix 4.3) raises concerns about reliability in safety-critical contexts. LLMs can exhibit sycophancy or hallucinate evidence when evaluating complex semantic criteria. While the authors report low disagreement rates with human experts (Table 10), the specific risks of automated grading in legal or medical contexts (e.g., missing a critical safety constraint) should be addressed more thoroughly. The paper should clarify the extent of human audit and the specific safeguards in place to prevent the propagation of grading errors in these sensitive domains.

Action items.

- [**writing**] The manuscript addresses a critical area in AI safety: the evaluation of proactive agents that infer hidden user intents. The separation of “Proactivity” (intent resolution) from “Completeness” (task success) is a valuable contribution for diagnosing agent behavior. However, from a safety and ethics perspective, there are significant concerns regarding the evaluation methodology and the handling of high-stakes domains. First, the evaluation of “hidden intents” relies entirely on a simulated user

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a well-structured benchmark for proactive agents, but the scientific evidence supporting the robustness of the metrics and the statistical significance of the findings requires strengthening.

First, the evaluation reliability analysis (Appendix E001, Table 1) is methodologically circular. The authors use GPT-5.4 as the primary grader for the main experiments and then audit its reliability by comparing it to itself and other LLMs (Claude, Gemini), with human experts serving only as a secondary check. To establish the validity of the Proactivity and Completeness scores, the ground truth for the 120 sampled trajectories must be established by human experts first. The current setup only measures inter-model agreement, not accuracy against a human-defined standard. If the LLM grader has a systematic bias (e.g., over-counting “inferred” intents), the agreement metrics will be high even if the scores are wrong.

Second, the sample size of 100 tasks (20 per persona) is on the lower end for benchmarking complex, long-horizon workflows. While the authors report standard deviations (e.g., 2.1% for GPT-5.4), the small number of tasks per domain makes it difficult to generalize performance differences across specific domains (e.g., the claim that “Law Trainee” tasks are hardest). The variance observed suggests that a single task type could disproportionately influence the average.

The authors should discuss the statistical power of their sample size or provide confidence intervals for the mean scores to demonstrate that the observed differences between models (e.g., GPT-5.4 vs. Qwen3.6 Plus) are statistically significant and not due to random task selection.

Finally, the ablation study (Section 4.3, Fig. 4) claims that removing prior sessions reduces Proactivity by an average of 9.5 points. However, the text lacks a statistical test (e.g., paired t-test) to confirm this drop is significant. Given the variance in the main results, a 9.5-point drop could potentially be within the noise margin without formal testing. The authors should report p-values or confidence intervals for this ablation result to substantiate the claim that prior interaction is a critical factor.

Action items.

- [**science**] The reliability audit (App. E001) uses GPT-5.4 as the primary grader for both Proactivity and Completeness, then validates against itself and other models. This introduces circularity; human experts should be the primary ground truth for the 120 sampled trajectories to validate the LLM grader's accuracy, not just agreement with other LLMs.
- [**science**] The sample size of 100 tasks (20 per persona) is relatively small for drawing robust conclusions about model performance across diverse domains (e.g., Law vs. Pharmacist). The reported standard deviations (e.g., 2.1% for GPT-5.4 Proc) suggest high variance; the paper should clarify if the 3 runs per task are sufficient to stabilize these estimates or if more tasks/runs are needed for statistical significance.
- [**science**] The ablation study (Fig. 4) claims a 9.5-point drop in Proactivity when removing prior sessions, but the text does not specify the statistical test used to confirm this difference is significant versus noise, nor does it report confidence intervals for the delta.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical analysis in this manuscript is generally descriptive, relying heavily on point estimates and standard deviations without rigorous hypothesis testing. While the separation of Proactivity (Proc) and Completeness (Comp) is conceptually sound, the statistical validation of the reported differences between models is insufficient for a benchmark paper making comparative claims.

First, Table 1 presents average scores with subscripts denoting standard deviations across three runs. However, the manuscript does not state whether the observed differences between models (e.g., GPT-5.4's 67.0 vs. Qwen3.6 Plus's 64.0) are statistically significant. With only 100 tasks and 3 repetitions, the effective sample size for model comparison is limited. The authors should perform appropriate statistical tests (e.g., paired t-tests or non-parametric equivalents like Wilcoxon signed-rank tests given the likely non-normal distribution of task scores) and report p-values. Furthermore, with 9 models and 2 metrics, multiple comparisons are inevitable; a correction method (e.g., Bonferroni or Benjamini-Hochberg) should be applied to avoid Type I errors when claiming one model "leads" or is "competitive."

Second, the reliability audit in Appendix E (Table 2) reports disagreement rates (e.g., 2.66% for human experts) but provides no measure of uncertainty. For a sample of 120 trajectories, the margin of error for a 2.66% rate is substantial. The authors should calculate and report 95% confidence intervals for these disagreement rates. Additionally, reporting an inter-rater reliability metric such as Cohen's Kappa or Krippendorff's Alpha would be more robust than simple disagreement percentages, as it accounts for chance agreement.

Third, the ablation study (Figure 4) claims that removing prior sessions reduces Proactivity by an average of 9.5 points. The text does not specify the variance of this difference or the statistical test used to confirm this drop is not due to random noise. Given the visual nature of the figure and the lack of error bars or significance markers, this claim remains anecdotal. A paired statistical test comparing the "with history" vs. "without history" conditions for the same tasks is required to substantiate the "markedly lowers" assertion.

Finally, the definition of Proactivity as a ratio of resolved intents assumes a uniform distribution of intent difficulty across tasks. If some tasks have significantly more hidden intents than others, the aggregate score could be biased. The authors should verify that the number of hidden intents per task is balanced across the 100 tasks or use a weighted average if the distribution is skewed.

Action items.

- [**science**] The paper reports standard deviations for aggregate scores (e.g., Table 1) but does not specify the statistical test used to determine significance between model rankings. Given the small sample size (N=100 tasks, 3 runs), authors must clarify if differences are statistically significant or merely descriptive, and address multiple-comparisons correction if claiming superiority.
- [**science**] The reliability audit (Table 2) reports disagreement rates but lacks confidence intervals or inter-rater reliability metrics (e.g., Cohen's Kappa). With only 120 sampled trajectories, the precision of these estimates is unclear; please provide 95% CIs for the disagreement rates to support claims of 'strong agreement'.
- [**science**] The ablation study (Fig. 4) claims a 9.5-point drop in Proactivity but does not report the variance or statistical significance of this difference. Without error bars or a paired test (e.g., Wilcoxon signed-rank) across the dependency groups, the magnitude of the effect cannot be rigorously validated.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_writing_quality.md`

The manuscript demonstrates a high level of technical clarity, with well-structured arguments and a logical flow from problem definition to experimental validation. The distinction between "Proactivity" and "Completeness" is articulated effectively, and the use of case studies in the appendix significantly aids readability by grounding abstract metrics in concrete examples.

However, there are minor issues with sentence-level precision and parallelism that slightly impede the smoothness of the reading experience. In Section 3, the description of the benchmark scale ("episode of 20 sessions, yielding 100 multi-turn tasks") could be misinterpreted by a reader unfamiliar with the specific experimental design; a brief clarification on the relationship between sessions and tasks would eliminate ambiguity. Additionally, in Section 4.2, the summary of model performance lacks strict parallel structure, which creates a slight rhythmic stumble in an otherwise polished paragraph.

The appendices, while dense, are generally well-organized. However, the explanation of the statistical difference between the aggregate intent distribution (Table A.2) and the reported proactivity score is currently too vague. The text states they are "different" without explicitly explaining the mathematical distinction (aggregate counts vs. averaged ratios), which is a crucial nuance for reproducibility. Finally, the failure analysis list in Appendix A.4, while clear, would benefit from a stricter check on verb consistency to ensure the list items feel like a cohesive set of observations rather than a collection of independent notes.

Overall, the writing is strong and professional, but these specific refinements would elevate the manuscript to a higher standard of clarity.

Action items.

- [**writing**] In Section 3 (Benchmark), the text states 'Each of the five user roles... has an episode of 20 sessions, yielding 100 multi-turn tasks.' This phrasing is slightly ambiguous regarding whether the 100 tasks are the sum of all sessions or a subset. Clarify if the 100 tasks are distinct from the 100 sessions (5 roles * 20 sessions) or if they are identical.
- [**writing**] In Section 4.2 (Main Results), the sentence 'GPT-5.4 leads on Proc, Claude 4.6 Opus on Comp, and Qwen3.6 Plus is competitive on both' lacks parallel structure. Consider revising to 'GPT-5.4 leads in Proc, Claude 4.6 Opus in Comp, and Qwen3.6 Plus is competitive in both' for better flow.
- [**writing**] In Appendix A.2 (Terminal Status of Hidden Intents), the phrase 'This distribution is different from the reported proactivity score' is vague. Specify that the reported score is an average of per-task ratios, whereas this table shows the aggregate distribution of intent statuses across all tasks.
- [**writing**] In Appendix A.4 (Failure Analysis), the list of failure patterns uses inconsistent verb forms: 'Ignoring...', 'Completing...', 'Failing...', and 'Using...'. While grammatically acceptable as a list of gerunds, ensure the subsequent descriptions maintain a consistent tense and voice (e.g., 'Agents treat...' vs 'Agents produce...').